

PSA:

- MATLABs ARE NOT GRADED BY ME SO IT TAKES A BIT.

- HW #1, Q10. b)

→ IGNORE INTEGRATING FACTOR / UNDETERMINED COEFFS.

THIS WAS SUPPOSED TO BE Q1 ON HW2, 😓 ops.

◦ You can do it after watching Preclass 4.

- EULER'S METHOD ERROR IS ON SECTION 7.1.

- TODAY'S LECTURE: HW 1, Q11-15.

◦ HOMEWORK 1 IS DUE BY TUESDAY 23:59 PM EST.

- STRUCTURE: GOAL.

555-625 pm Quiz

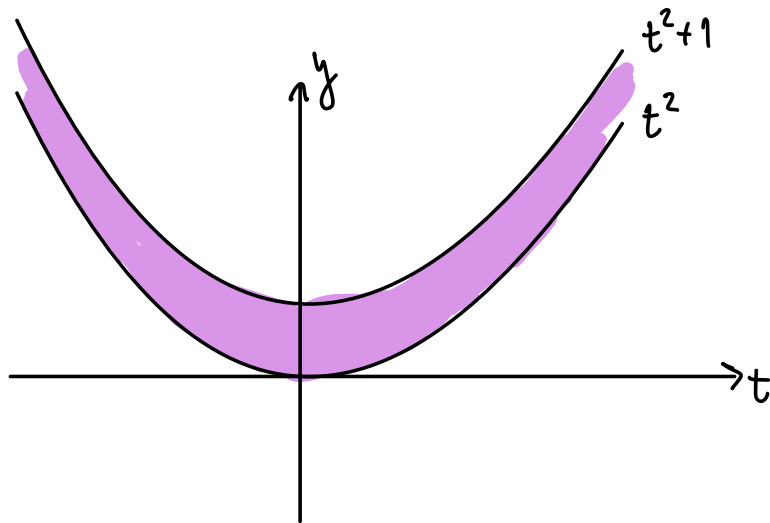
630-730 pm Lecture pt 1.

730-745 pm stretching/water/bathroom break.

745-845 pm Lecture pt 2.

Quiz 3.

Q2b) Check conditions for T.E.U.



Lecture 3. 1.6 Equilibria and the Phase Line (Preclass)
 1.7 Bifurcations
 1.8 Linear Equations.

◦ What kind of DE are phase lines useful for?

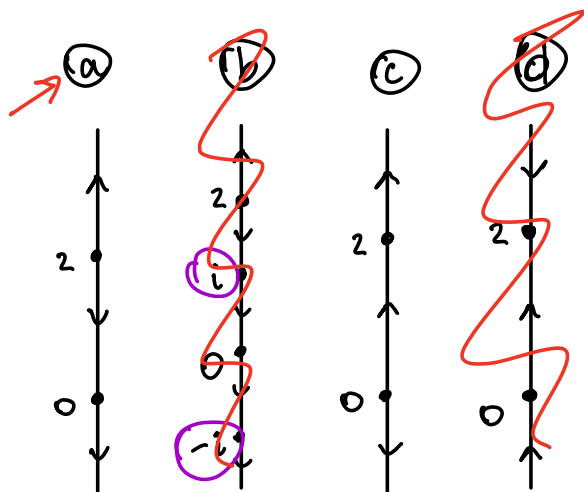
• autonomous equations, $y' = f(y)$

◦ What are the three types of equilibria?

- source (repels)
- sink (attract)
- node (indiff.)

(3 min-poll)

Ex: What is the phase line of $\frac{dy}{dt} = \underbrace{(y^2+1)}_{>0} \underbrace{(y-2)}_{>0} y^2$



equilibria: 0, 2

$$y=3: \frac{dy}{dt} = \underbrace{(\quad)}_{>0} \underbrace{(3-2)}_{>0} \cdot \underbrace{3^2}_{>0}$$

Ex: Modified logistic model

k : growth rate

N : carrying capacity

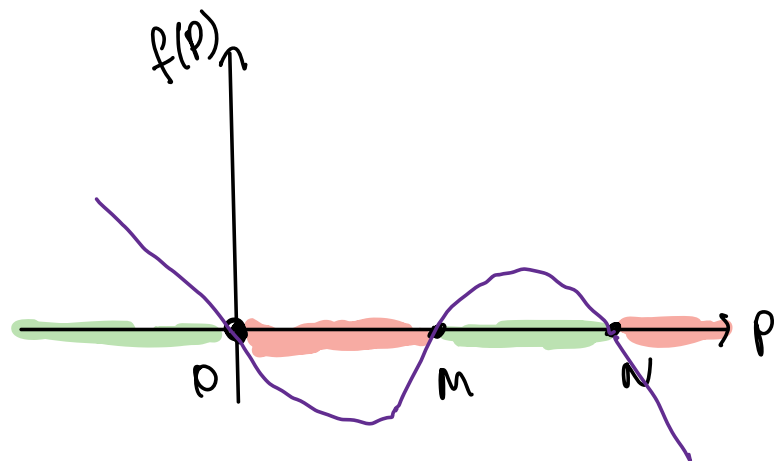
M : sparsity constant

$M < N$

$$P' = kP \left(1 - \frac{P}{N}\right) \left(\frac{P}{M} - 1\right)$$

$$kP \cdot \left(-\frac{P}{N}\right) \left(\frac{P}{M}\right) \sim C \cdot (-P^3)$$

- Draw the phase-line. Pro-tip: Draw the graph of $f(P)$.



- Space for general questions or remarks.

BIFURCATIONS

When we introduce a parameter to a diff. eq., we may write it as

$$\frac{dy}{dt} = f(t, y, \lambda) \quad \text{for some } \lambda \in \mathbb{R}.$$

Parameters we did so far:

- examples {
- growth rate k
 - population capacity N
 - sparsity constant M
 - predator-prey constants: A, B, C, D
 - gravitational force: g
 - drag coefficient: k (usually β or b)

Our focus: change 1-parameter at each time.

"def". A bifurcation point $\lambda_0 \in \mathbb{R}$ is a value that the qualitative information of nearby points is NOT the same.

For example:

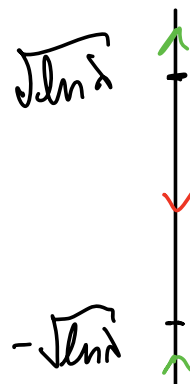
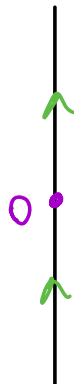
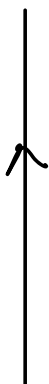
Bif: $y' = e^{y^2} - \lambda$

$\lambda < 0$

$0 < \lambda < 1$

$\lambda = 1$

$\lambda > 1$



$e^{y^2} - \lambda = 0$
 $e^{y^2} = \lambda$

$y = \pm \sqrt{\ln \lambda}$

Not bif: $y' = y - \lambda$

$y = \lambda$

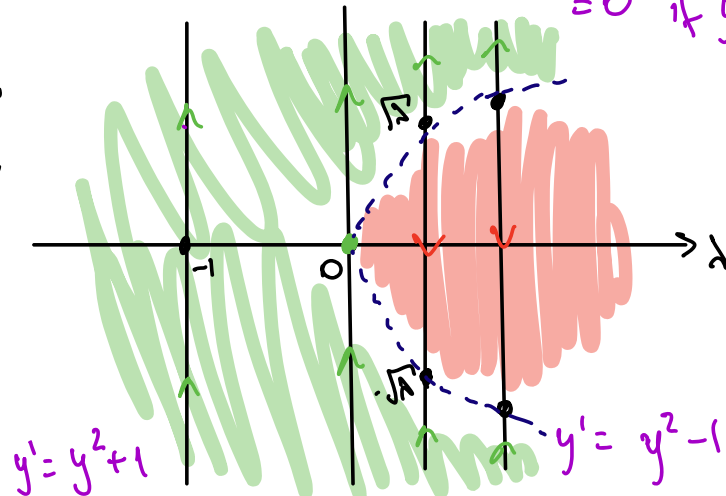


Claim: equilibrium solutions are key to understand bifurcations.

Ex: $\frac{dy}{dt} = y^2 - \lambda$

$\lambda = 0 \Rightarrow y' = y^2 > 0$ if $y \neq 0$
 $= 0$ if $y = 0$.

Bifurcation
Diagram

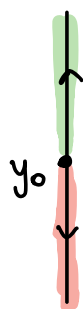


How exactly do we track that kind of change?

• Info on equilibria:

Suppose $\frac{dy}{dt} = f(y)$, f smooth function

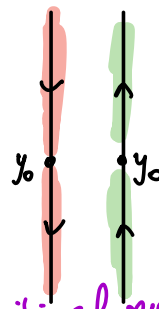
Q: What is the value of $f'(y_0)$? y_0 is equilibria, $f(y_0) = 0$.



f is increasing
source: $f'(y_0) > 0$



f is dec.
sink: $f'(y_0) < 0$

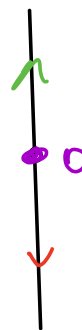
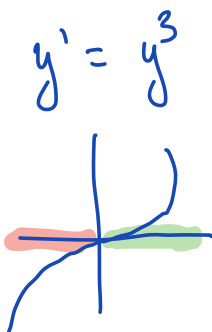


critical point of f
node: $f'(y_0) = 0$.

Poll

Q: if $f(y_0) = 0$, does $f'(y_0) = 0$ imply a node?

- Yes
- ① No



Linearization theorem

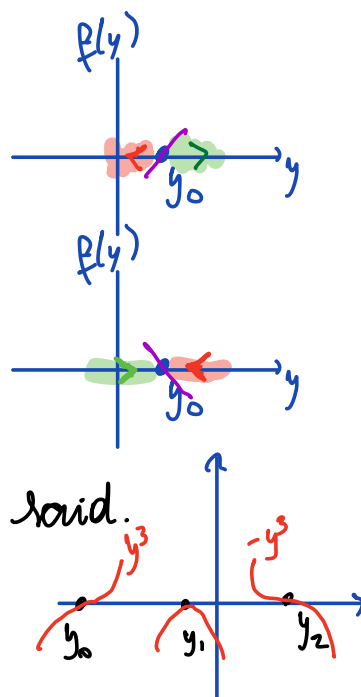
Consider the ODE

$$\frac{dy}{dt} = f(y)$$

for f continuously differentiable. (f' is continuous)

Suppose $f(y_0) = 0$.

- if $f'(y_0) > 0$, y_0 is a source.
- if $f'(y_0) < 0$, y_0 is a sink.
- if $f'(y_0) = 0$, nothing can be said.



Ex: $y' = (y+1)^2(y-1)(y-2)^3 = f(y)$

$$f'(y) = 2(y+1)(y-1)(y-2)^3 + (y+1)^2 \cdot 1 \cdot (y-2)^3 + (y+1)^2(y-1) \cdot 3(y-2)^2$$

$$f'(-1) = 0$$

$$f'(1) = 0 + (1+1)^2 \cdot 1 + (1-2)^3 + 0 = 4 \cdot (-1) = -4.$$

$y = 1$ is a sink.

Example with parameters:

- logistic model with harvesting.

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) - C \quad , \text{ fix } k > 0, N > 0$$

$$kP - \frac{k}{N}P^2 - C = 0$$

C is free, but > 0 .

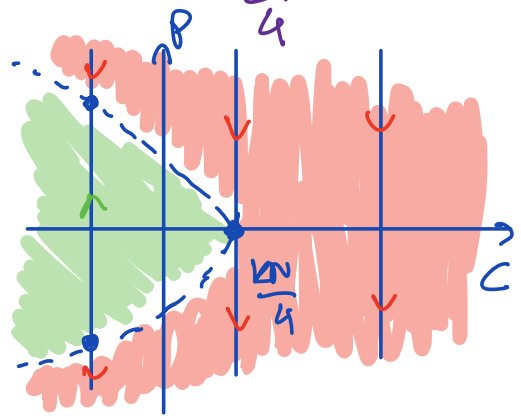
- Equilibria: $f(P) = 0$ ↘ quadratic formula

$$f(P) = 0 \Leftrightarrow \frac{N}{2} \pm \sqrt{\frac{N^2}{4} - \frac{CN}{k}} \leadsto C \leq \frac{kN}{4}$$

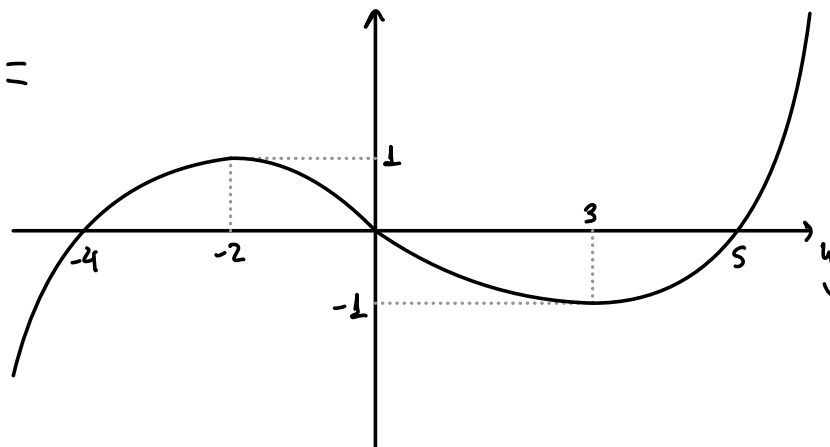
$C < \frac{kN}{4}$: 2 equilibria

$C = \frac{kN}{4}$: 1 equilibrium

$C > \frac{kN}{4}$: 0 equilibrium



Ex: let $f(y) =$



poll

Consider $y' = f(y) + \beta$. How many bif. points are there for $\beta \in \mathbb{R}$?

a) 0

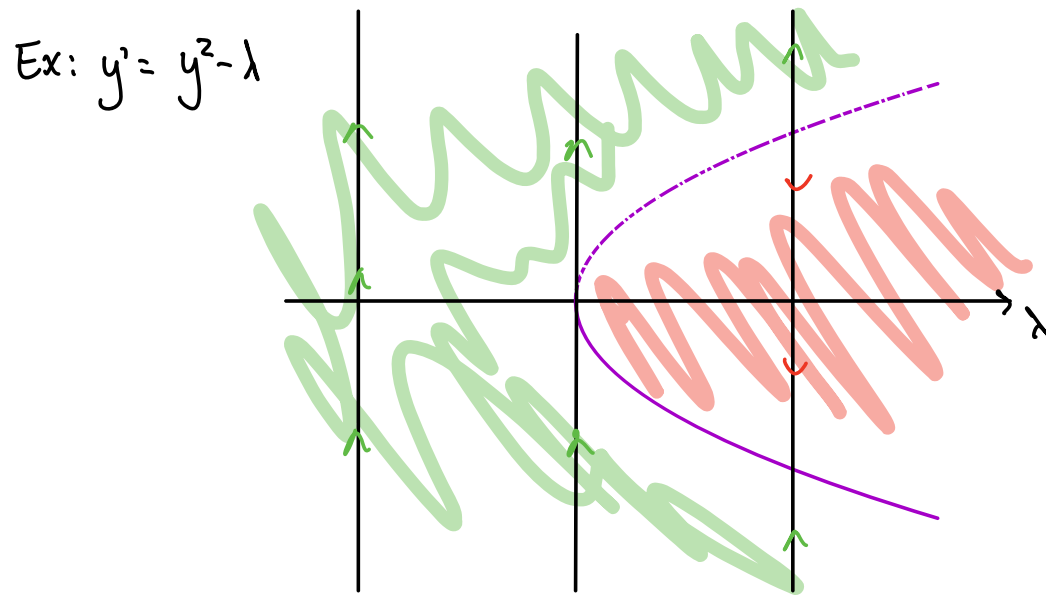
b) 1

~~c) 2~~

d) 3

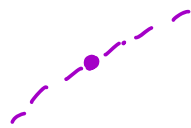
$\beta = \pm 1.$

Bifurcation Diagram: Draw a collection of phase-lines.



Remark on "hyperbolicity".

• if source,



$$f'(y_0) > 0$$

• if sink,



$$f'(y_0) < 0$$

Ex. Draw the bifurcation diagram for

$$\frac{dy}{dt} = y^2 - ky + 9, \quad k \in \mathbb{R}.$$

homework.

Linear Systems.

- What does linearity even means? (operator)

$$f(x) = ax + b$$

$$T: \mathbb{R} \rightarrow \mathbb{R}$$

Linear eq.

$$y' = f(t, y) = a(t) \cdot y + b(t)$$

$$T(x+y) = T(x) + T(y)$$

$$T(rx) = r T(x)$$

$L(\text{function}) =$ how the derivative behaves.

$$L[y] = \frac{dy}{dt} = f(t, y)$$

- Homogeneous vs Non-homogeneous

$$b(t) = 0$$

$$b(t) \neq 0$$

Recap: $\frac{dy}{dt} = \overset{t^2}{a(t)} \underset{e^{\text{test}}}{y} + b(t)$

a line in y

Classify the DE by order, linearity, homog., separable.

DE	order	linear	homog.	sep.
$\frac{dy}{dt} = \frac{1}{2y+1}$	1st	X	X	✓
$y' + \underbrace{e^t}_{a(t)} y = \underbrace{\cos(t)}_{b(t)}$	1st	✓	X	X
$y'' + y' + y = 0$	2nd	✓	✓	—
$y' + e^y = 0$ $\uparrow$ $y' = e^y$ $\frac{y'}{e^y} = -1$	1st	X	✓	✓
$y' + \frac{y}{t} = \frac{1}{t^2}$ $a(t) = \frac{1}{t}$ $b(t) = \frac{1}{t^2}$	1st	✓	X	X

Linearity Principle:

⊛ A homogeneous linear equation has the following property:

$y' = a(t)y$ and has solutions $y_1(t), y_2(t)$

then $c_1 y_1 + c_2 y_2$ is also a solution.

Proof: $L[y_1] = a(t)y_1 = 0^*$

$L[y_2] = a(t)y_2 = 0^*$

$$L[c_1 y_1 + c_2 y_2] = c_1 \underbrace{L[y_1]}_0 + c_2 \underbrace{L[y_2]}_0 = 0.$$

Extended linearity principle:

Let $y' = a(t)y + b(t)$ be a first order linear ODE.

then the gen. sol

$$y(t) = y_h(t) + y_p(t)$$

where $y_h(t)$ is a solution for the homogeneous part

$y' = a(t)y$ and $y_p(t)$ is a particular solution.

for the ODE.

Proof: Let y be any solution.

$$y_h = y - y_p : \quad (y - y_p)' = a(t)y + \cancel{b(t)} - [a(t)y_p + \cancel{b(t)}]$$
$$= a(t)[y - y_p]$$

$$y(t) = y_h + y_p.$$

Ex: $y' = \frac{2t}{t^2+1}y + \frac{4t}{t^2+1}$ Hint: $y_p = 2$ is a solution.

What is a homogeneous solution like?

• $y' = \frac{2t}{t^2+1}y \leadsto \int \frac{dy}{y} = \int \frac{2t}{t^2+1} dt$

$$\ln|y| = \ln|t^2+1| + C$$

$$y_h = C \cdot (t^2+1) = Ct^2 + C$$

the general solution is given by

$$y(t) = y_h + y_p = Ct^2 + C + 2.$$

Ex. $y' = (t-3)y - 2t^2 + 6t + 2$

Hint: $y(t) = 2t$
is a solution.

Homogeneous part:

$$y' = (t-3)y \quad \Rightarrow \quad \frac{dy}{y} = (t-3)dt$$

$$\ln|y| = \frac{t^2}{2} - 3t + C.$$

$$y_h(t) = C \cdot e^{\frac{t^2}{2} - 3t}$$

the general solution is given by

$$y(t) = C \cdot e^{\frac{t^2}{2} - 3t} + 2t$$

Ex: $\frac{dy}{dt} = y^2$ has solutions → not linear.

Poll-1min $y_1(t) = \frac{1}{1-t}$ and $y_2(t) = \frac{2}{1-t}$

Does this contradict the linearity principle?

• is $y_1 + y_2$ also a solution?

$$(y_1 + y_2)' = \left(\frac{3}{1-t}\right)' = 3 \cdot \frac{1}{(1-t)^2} \neq (y_1 + y_2)^2 = \frac{9}{(1-t)^2}$$

A: It doesn't contradict. the ODE is not linear, so the principle doesn't even apply.

Ex: $\frac{dy}{dt} = -y + 2$ has solutions
 $b(t) \neq 0$

Poll-1min $y_1(t) = 2$ and $y_2(t) = 2 - e^{-t}$

Is the Linearity Principle valid in this case?

Q: Is $y_1 + y_2$ a solution?

A: $y_1 + y_2 = 4 - e^{-t}$

$$(y_1 + y_2)' = -e^{-t} \neq -(4 - e^{-t}) + 2 = e^{-t} - 2.$$

A: It is not valid, because the equation is not homogeneous.

How to solve 1st order linear diff. equations?

- INTEGRATING FACTOR
- UNDETERMINED COEFFICIENTS.

— TO BE CONTINUED : PRECLASS 4. (15 min)

Reminder: office hours Wed are after class.